

Selecting an EDA Tool for Schematic Capture, Layout and Simulation

Electronic Design Automation (EDA) tools are used worldwide to design electronic systems. These tools are a boon to designers and researchers, because they enhance productivity by using the capabilities of a computer to assist them in creating, modifying, analysing and optimising electronic systems. This is one tool that has to be consistent throughout the organisation for better communication between departments, which makes choosing the correct tool even more important. Here is what you should be looking at...

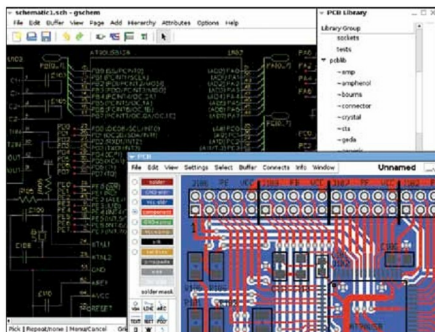


Fig. 1: An EDA tool

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Electronic design automation (EDA or ECAD) is a category of software tool used to create, modify, analyse and optimise the designs of electronic systems. These tools assist the designer throughout the product design cycle and automate a lot of standard tasks to save time and energy, freeing designers to address more important challenges. EDA software increases the productivity of the designer, improves the quality of design, improves communication

through documentation, and generates industry-standard output data directly for manufacturing.

How was it done in the old days?

Before the idea of EDA was conceived, the designs were laid out by hand. Some expert designers used geometric software to make layouts. The process was fundamentally graphic, with the translation from electronics to graphics done manually. By the mid-1970s, designers started automating the drafting of

designs when tools for placement and routing came into existence.

But the major breakthrough occurred in 1980 with the publication of 'Introduction to VLSI Systems' by Carver Mead and Lynn Conway. This ground-breaking text advocated chip design with programming languages. This idea of using a programming language to define electronic components is the core of current-world EDA tools. The immediate result was a considerable increase in the complexity of the chips that could be designed, with improved access to design verification tools that used logic simulation.

How EDA tools increase productivity?

EDA tools increase productivity multi-fold. These tools support every phase of the product design cycle and make each task more organised, traceable and quick. When you want to design a new project, you start with making the schematic. Now, if you were to do this by hand you would have to draw each symbol manually, referring to the datasheet. You cannot remember the pin configuration of each component and most of the design cannot be completed in one day. This means that every time you come back to the project, you will be referring to the datasheets repeatedly for each component to draw symbols. Isn't this cumbersome?

The EDA tools simplify such repetitive tasks with the help of a com-